

Primary Mirror Thermal Control

Prediction Methods and Control Algorithms for Rubin Observatory

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Analysis performed with extensive use of Claude Code

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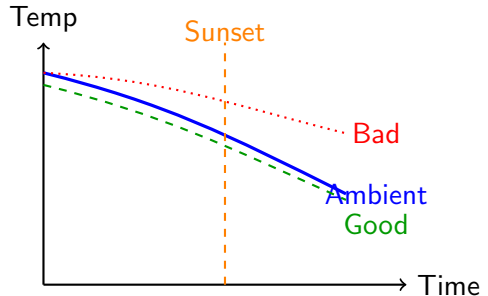
The Challenge

At sunset, the mirror must:

- Match ambient temperature
- Match ambient cooling rate
- Preferably be slightly cold

Why it's hard:

- 2-hour thermal time constant
- Must predict 3+ hours ahead
- Weather is variable



Good control anticipates the cooling trend

Two Key Questions

1. Prediction: What will conditions be at sunset?

- Temperature T_{sunset} : How cold will it be?
- Rate $\dot{T}_{\text{sunset}}$: How fast is it cooling?

2. Control: How do we get the mirror there?

- Pre-sunset: Drive to predicted conditions
- Overnight: Track ambient as it evolves

Key Insight

Matching both temperature **AND** rate at sunset dramatically improves overnight performance.

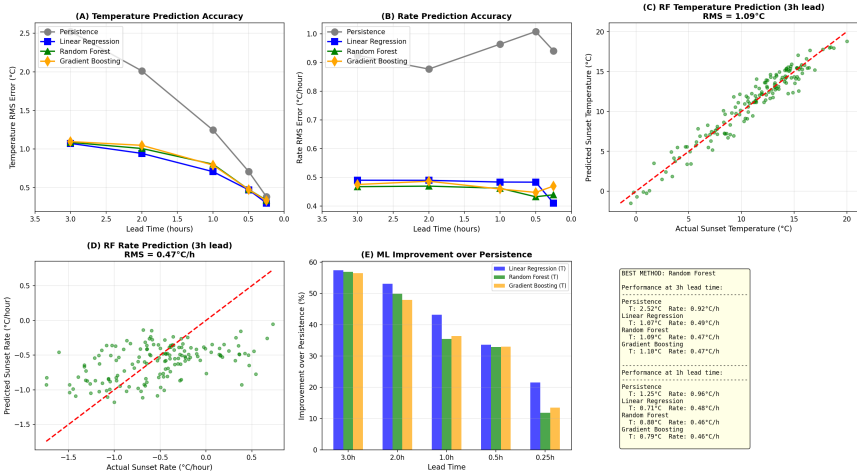
Prediction Methods Tested

Method	Type	Description
Ridge Regression	Traditional ML	Simple, fast, interpretable
Random Forest	Ensemble ML	Best overall performance
Gradient Boosting	Ensemble ML	Competitive with RF
LSTM Network	Deep Learning	Sequence modeling
Transformer	Deep Learning	Attention mechanism

Features used:

- Recent temperature history (3 days)
- Seasonal encoding (day of year)
- Previous days' sunset temperatures

Prediction Results: Temperature



At 3-hour lead time:

- Random Forest: 1.09°C RMS (best)

LLM-Inspired Approaches: Can We Beat Random Forest?

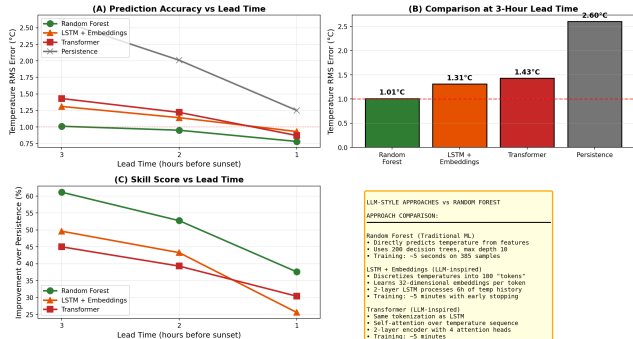
The Idea: Treat Temps as “Text”

- 1 Tokenize temps into 100 bins
- 2 Learn embeddings (like word vectors)
- 3 Process with LSTM or Transformer
- 4 Predict temperature change

Results at 3h Lead Time:

Method	RMS
Random Forest	1.01°C
LSTM + Embed	1.31°C
Transformer	1.43°C

Temperature Prediction: Traditional ML vs LLM-Inspired Approaches



LLM-STYLE APPROACHES vs RANDOM FOREST

APPROACH COMPARISON:

- Random Forest (Traditional ML)**
- Directly predicts temperature from features
 - Uses 200 decision trees, max depth 10
 - Training: ~5 seconds on 385 samples
- LSTM + Embeddings (LLM-Inspired)**
- Discretizes temperatures into 100 “tokens”
 - Learns 32-dimensional embeddings per token
 - 2-layer LSTM processes 6h of temp history
 - Training: ~5 minutes with early stopping
- Transformer (LLM-Inspired)**
- Same tokenization as LSTM
 - Self-attention over temperature sequence
 - 2-layer encoder with 4 attention heads
 - Training: ~5 minutes

RESULTS AT 3-HOUR LEAD TIME:

Random Forest:	1.01°C RMS (61% vs persist)
LSTM + Embeddings:	1.31°C RMS (56% vs persist)
Transformer:	1.43°C RMS (45% vs persist)
Persistence:	2.60°C RMS (baseline)

WHY RF OUTPERFORMS NEURAL NETWORKS:

1. Limited data (385 training samples)
2. Tabular data favors tree methods
3. No clear sequential dependencies
4. Tokenization loses temperature precision

Why RF Wins

Rate Prediction Accuracy

Method	RMS ($^{\circ}\text{C}/\text{h}$)
Ridge Regression	0.49
Random Forest	0.47
Gradient Boosting	0.47

Key Finding:

Rate prediction error is **constant across lead times** ($\sim 0.47^{\circ}\text{C}/\text{h}$).

This means:

- Atmospheric variability is the limit
- Earlier predictions aren't worse
- We can confidently use 3h predictions

Use Random Forest for Both Predictions

	Temperature	Rate
At 3h lead	1.09°C RMS	0.47°C/h RMS
At 1h lead	0.80°C RMS	0.46°C/h RMS

Why Random Forest?

- Best accuracy for both quantities
- Robust to missing data
- Fast inference (<10ms)
- Easy to retrain monthly

Control Algorithm: The Key Formula

Rate-Matching Setpoint

$$T_{\text{setpoint}} = T_{\text{target}} + \tau \cdot \dot{T}_{\text{target}} - \sigma_{\text{cold}}$$

- T_{target} : Target temperature (predicted or measured)
- $\dot{T}_{\text{target}}$: Target cooling rate (typically negative)
- $\tau = 2$ hours: Mirror thermal time constant
- σ_{cold} : Cold bias (0.2°C pre-sunset, 0.1°C overnight)

Physical meaning: If ambient is cooling at 1°C/h, set mirror 2°C colder so it warms toward ambient at the same rate.

Critical constraint: Setpoint rate limited to $\pm 2^\circ\text{C}/\text{hour}$ to prevent thermal gradients.

Control Algorithm: Dual-Setpoint Approach

Pre-Sunset (dome closed)

- HVAC holds interior at **FLAT** temp
- Mirror equilibrates naturally
- Apply 0.3°C cold bias

$$T_{HVAC} = T_{pred} - 0.3C \text{ (constant)}$$

No tracking of exterior temp before sunset

At Sunset

Dome opens $\rightarrow$ interior air instantly = exterior

Overnight (dome open)

- Rate-matching control
- 0.3°C cold bias

$$T_{sp} = T_{amb} + 2h \cdot \dot{T}_{amb} - 0.3C$$

Control Results: At Sunset

Metric	Temp-Only	Rate-Matching
Temperature RMS	0.80°C	0.69°C
Rate RMS	0.87°C/h	0.34°C/h
% Days Warm	65%	40%

Key Result

Rate-matching reduces rate error by **61%** and achieves desired cold bias.

Overnight Tracking Error

Strategy	RMS
Temperature-Only	0.53°C
Rate-Matching	0.30°C
<i>Improvement</i>	<i>44%</i>

Why it works:

Better initial conditions (matching both T and $\dot{T}$) propagate through the entire night.

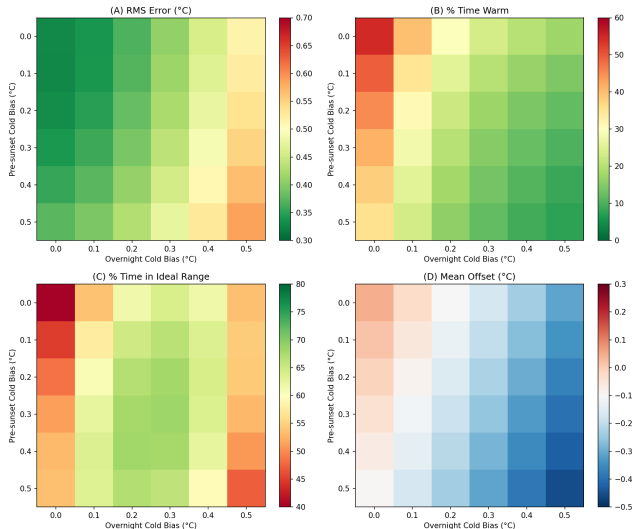
At sunset: Mirror is already tracking the ambient trend, not fighting against it.

Cold Bias Optimization

Finding the Optimal Cold Bias

Bias	RMS	% Warm
0.0°C	0.33°C	55%
0.1°C	0.34°C	35%
0.2°C	0.38°C	22%
0.3°C	0.44°C	15%
0.4°C	0.51°C	11%

0.3°C gives 68% ideal range, only 15% warm.



Adaptive vs Fixed Bias

Should bias adapt to conditions?

Tested approaches:

- Rate-adaptive: $\sigma = 0.15 + k|\dot{T}|$
- Uncertainty-adaptive
- Combined

Strategy	% Warm	% Ideal
Fixed 0.2°C	17%	72%
Fixed 0.3°C	11%	70%
Rate-Adaptive	15%	63%

Conclusion:

Fixed bias wins!

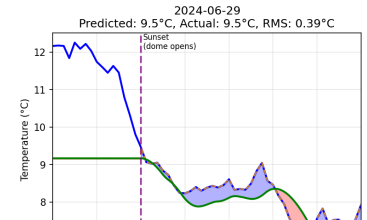
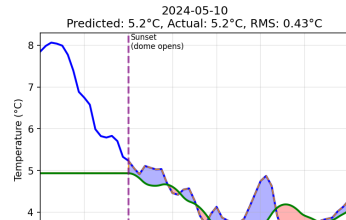
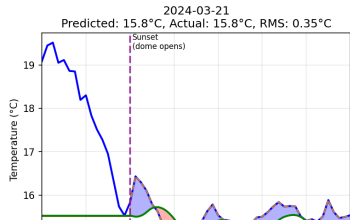
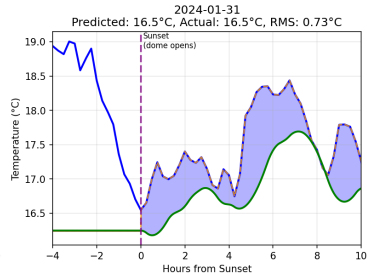
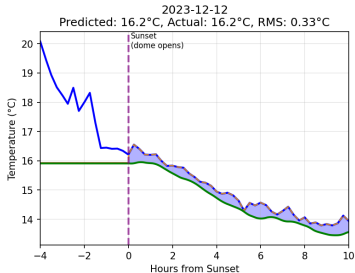
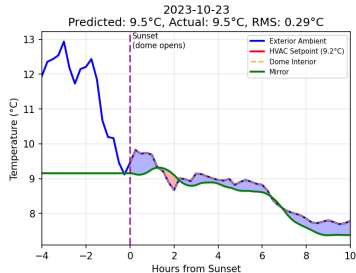
- Simpler to implement
- More robust
- Adaptive adds complexity without clear benefit

Recommended: 0.3°C fixed bias

- 70% ideal range
- Only 11% warm
- Mean offset -0.31°C

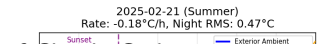
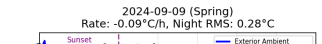
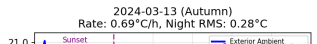
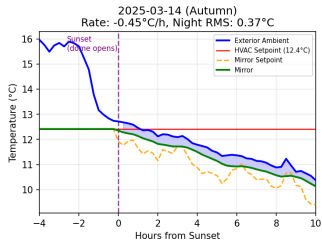
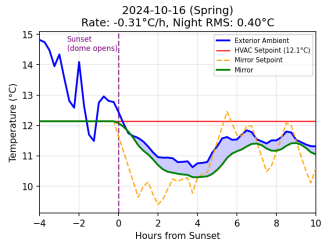
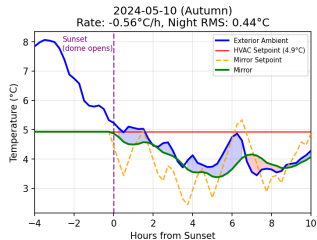
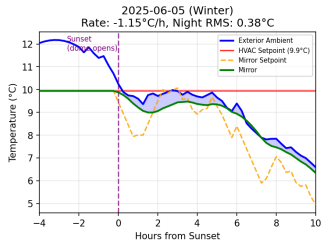
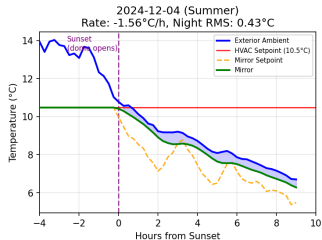
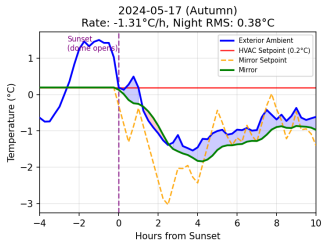
Dual-Setpoint Control Examples

FIXED Model: Mirror at HVAC Setpoint Before Sunset (FLAT, not tracking exterior)
Pre-sunset bias=0.3°C, Overnight bias=0.3°C



Example Day Operation

CORRECTED Model: Mirror at FLAT HVAC Setpoint Before Sunset Pre-sunset bias = 0.3°C, Overnight bias = 0.3°C



Summary: Key Numbers

Metric	Value
<i>Prediction (3h lead)</i>	
Temperature RMS	1.09°C
Rate RMS	0.47°C/h
<i>Dual-Setpoint Control (0.3°C bias)</i>	
Overnight Tracking RMS	0.44°C
Mean Offset	−0.26°C (cold)
% Time in Ideal Range	68%
% Time Warm	15%

Recommendations

1. Use Random Forest Predictions

Train separate models for temperature and rate. Retrain monthly.

2. Dual-Setpoint Control

Pre-sunset: $T_{HVAC} = T_{pred} - 0.3C$ (FLAT, constant)

Overnight: $T_{sp} = T_{amb} + 2h \times \dot{T}_{amb} - 0.3C$

3. Rate Limit Setpoint Changes

Max $\pm 2C$ /hour to prevent thermal gradients.

Expected Outcome

0.44°C RMS overnight. Mean $-0.26C$ (few tenths cold). 68% ideal. Only 15% warm.

Thank you

Full Reports Available:

- Report 1: Prediction Methods Comparison (11 pages)
- Report 2: Thermal Control Algorithms (12 pages)

Thermal Control Analysis Team
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